

TELECOM CUSTOMER CHURN ANALYTICS REPORT

Churn Prediction · CLTV Analysis · Customer Segmentation · Executive Dashboard

IBM Telco Customer Churn Dataset | 7,043 Customers

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1. Executive Summary

This report presents a comprehensive customer analytics study for a telecommunications company, using the IBM Telco Customer Churn Dataset comprising 7,043 customer records. The project was designed to answer critical business questions like why do customers leave, who is most at risk, what is the revenue exposure, and can machine learning predict future churn before it happens?

The analysis moves through seven structured phases of business understanding, exploratory data analysis, data cleaning, and feature engineering, churn analysis, machine learning modelling, customer segmentation, and executive reporting. Three ML classifiers were trained and compared, with Gradient Boosting emerging as the top performer. K-Means clustering identified four distinct customer segments, each with different churn profiles and retention priorities.

Key Performance Indicators

7,043 Total Customers IBM Telco Dataset	~26.5% Churn Rate Overall	4 Customer Segments K-Means Clusters	3 ML Models Built LR · RF · GB
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The project demonstrates how predictive analytics can help telecom companies proactively reduce customer churn and improve customer retention strategies.

2. Business Problem & Objectives

Customer churn is one of the most critical and costly challenges in the telecommunications industry. Acquiring a new customer typically costs five to seven times more than retaining an existing one. Every churned customer represents not only immediate lost revenue but also long-term lifetime value that is permanently removed from the business.

This project was structured to answer the following business questions:

- What is the overall churn rate, and how does it break down across customer groups?
- Which contract types, internet services, and payment methods are most associated with churn?
- What are the primary reasons customers give for leaving?
- Which customers generate the highest lifetime value (CLTV)?
- How much revenue is currently at risk due to churn?
- Can machine learning accurately predict future churn?
- Can customers be segmented into actionable groups for targeted retention?

3. Dataset Overview

The IBM Telco Customer Churn Dataset is a widely used benchmark dataset for customer retention modelling. It contains 7,043 customer records and 33 features covering four primary domains:

Domain	Features Included
Demographics	Age, Gender, Married, Senior Citizen, Dependents
Services	Internet Service, Internet Type, Phone Service, Streaming, Online Security
Billing	Contract Type, Payment Method, Paperless Billing, Monthly Charge, Total Charges, Total Revenue
Churn Signals	Churn Label, Churn Score, Churn Reason, CLTV, Tenure in Months

No missing values were detected in the dataset after initial quality checks, making it a clean and reliable foundation for modelling.

4. Churn Analysis & Business Insights

4.1 Overall Churn Distribution

The dataset reveals a churn rate of approximately 26.5%, meaning roughly one in four customers left the company. This is significantly above the industry benchmark of 15-20% for telecom providers, indicating a meaningful retention problem that demands proactive intervention.

4.2 Churn by Contract Type

Contract type was among the strongest churn predictors. The analysis revealed a stark contrast:

Contract Type	Churn Rate	Risk Level
Month-to-Month	Highest	Critical
One Year	Moderate	Medium
Two Year	Lowest	Low

Month-to-Month customers have no switching cost and represent the largest and most at-risk cohort. Converting even a fraction of this group to annual contracts would materially reduce churn.

4.3 Churn by Internet Type

Fiber Optic internet customers show a disproportionately high churn rate compared to DSL or no-internet customers. This pattern suggests either a price-sensitivity issue (Fiber is typically the premium tier) or an unmet expectation gap for customers paying more expect more, and when the experience falls short, they leave.

4.4 Revenue at Risk

Revenue at risk was quantified by summing Total Revenue across all churned customers in the dataset. This figure represents the historical revenue already lost and can be extrapolated using ML churn probabilities to estimate forward-looking risk for active customers.

5. Machine Learning Models

5.1 Modelling Approach

Three binary classification models were trained on an 80/20 train-test split with stratified sampling to handle class imbalance. Features included both numerical variables (Age, Tenure, Monthly Charge, Total Revenue, CLTV, Total Charges) and encoded categorical variables (Contract Type, Internet Type, Payment Method, Gender, Married, Dependents, Paperless Billing).

5.2 Model Performance Comparison

Model	Accuracy	ROC-AUC	Notes
Gradient Boosting	~81%	Best	Selected
Random Forest (300 trees)	~80%	Runner-up	Feature importance
Logistic Regression	~79%	Baseline	Interpretable

Gradient Boosting was selected as the final model based on its superior ROC-AUC score, which measures a model's ability to distinguish churners from non-churners across all probability thresholds a more meaningful metric than accuracy alone in imbalanced classification problems.

5.3 Top Churn Predictors (Feature Importance)

Random Forest feature importance analysis revealed the strongest signals for predicting churn:

#	Feature	Business Interpretation
1	Tenure in Months	Newer customers are far more likely to churn
2	CLTV	Low predicted lifetime value correlates with churn risk
3	Total Revenue	Historical revenue pattern is a strong retention signal
4	Monthly Charge	Higher bills increase churn sensitivity
5	Contract Type	Month-to-Month = highest churn risk
6	Total Charges	Cumulative spend history indicates relationship depth

6. Customer Segmentation

6.1 K-Means Clustering Approach

K-Means clustering was applied to segment the customer base using five behavioral and financial variables: Age, Tenure in Months, Monthly Charge, Total Revenue, and CLTV. The Elbow Method was used to determine the optimal number of clusters, which converged at 4 distinct segments. All features were standardized using StandardAero before clustering to prevent high-magnitude variables from dominating distance calculations.

6.2 Customer Segment Profiles

Cluster	Profile	Churn Risk	CLTV Tier	Retention Priority
0	New, low-spend	High	Low	Onboarding focus
1	Mid-tenure, moderate	Medium	Medium	Loyalty programmes
2	Long-tenure, high-value	Low	High	VIP retention
3	Senior, high charge	Medium-High	High	Personalized outreach

The top 10% of customers by CLTV were additionally flagged as a priority group, warranting dedicated account management and proactive retention before any churn signals emerge.

7. Customer Lifetime Value (CLTV) Analysis

Customer Lifetime Value is the single most important metric for prioritizing retention investment. Not all customers are equal, losing a high-CLTV customer is far more damaging than losing a low-value one, and the cost of retention efforts must be proportional to the value at stake.

The CLTV analysis revealed significant variation across customer segments and contract types. Key findings include:

- Annual and two-year contract holders consistently show higher CLTV than month-to-month subscribers
- High-CLTV customers tend to have longer tenure and lower monthly charge volatility
- The top 10% high-value customer cohort accounts for a disproportionate share of total company revenue
- CLTV is one of the top 5 machine learning predictors lower CLTV customers churn at higher rates

This insight directly informs retention investment allocation: the highest CLTV customers should receive proactive, personalized outreach rather than reactive engagement after churn signals appear.

8. Business Recommendations

Based on the churn analysis, ML modelling, and segmentation findings, the following strategic recommendations are proposed:

#	Action	Rationale
1	Contract upgrade campaigns	Month-to-Month customers are the single highest-risk group. Offering incentives to switch to annual contracts directly reduces churn probability.
2	Fiber Optic service improvement	Fiber customers churn at disproportionate rates despite paying premium prices. Investigate service quality, speed consistency, and support response times.
3	High-CLTV proactive retention	Top 10% CLTV customers should receive dedicated account management and proactive outreach before churning signals to appear.
4	New customer onboarding	Tenure is the strongest churn predictor. The first 6 months are the highest risk. Structured onboarding and early check-ins can significantly improve early retention.
5	Cluster-targeted campaigns	Use the 4 K-Means segments to tailor communication, offers, and incentives. Generic campaigns underperform personalized segment-level messaging.
6	Deploy ML scoring pipeline	Score all active customers monthly using the Gradient Boosting model. Flag high-probability churners for immediate retention team intervention.

9. Technical Overview

Category	Tools & Libraries
Language	Python 3.10+
Data Manipulation	Pandas, NumPy
Visualisation	Matplotlib, Seaborn, GridSpec
Machine Learning	Scikit-Learn Logistic Regression, RandomForestClassifier, GradientBoostingClassifier, KMeans, Standard Scaler, Label Encoder
Evaluation Metrics	Accuracy, Precision, Recall, F1 Score, ROC-AUC, Confusion Matrix, Classification Report
Environment	Google Collab / Jupyter Notebook

Notebook Structure

Section	Content
Section 0	Environment setup, library imports, reproducibility configuration (seed=42)
Section 1	Data loading from IBM Telco dataset, business framing and objectives
Section 2	Exploratory data analysis churn distribution, contract types, internet services, CLTV histograms
Section 3	Data cleaning, missing value handling, feature engineering, encoding, scaling
Section 4	Churn analysis by contract, internet type, payment method, churn reasons, revenue at risk
Section 5	ML modelling Logistic Regression, Random Forest (300 trees), Gradient Boosting, feature importance
Section 6	K-Means clustering (4 segments), cluster profiling, CLTV and revenue by segment, high-value customer flag
Section 7	Executive KPI dashboard, business recommendations, all charts saved as high-res PNG files

10. Conclusion

This project demonstrates how customer analytics can transform raw telecom data into targeted business action. By combining exploratory analysis, machine learning, and customer segmentation, the analysis delivers a clear picture of churn dynamics and a practical roadmap for reducing revenue loss.

The Gradient Boosting classifier provides a reliable, deployable mechanism to score existing customers by churn probability, enabling the retention team to intervene before customers decide to leave. The four-segment K-Means model ensures that retention resources are allocated where they will have the greatest impact rather than applied generically across the entire customer base.

The most important operational insight from this work is that churn is predictable and preventable. Tenure, CLTV, contract type, and monthly charge collectively explain the majority of churn risk — and all four are either directly influenceable (contract upgrades, pricing strategy) or early-warning signals that can trigger proactive retention action.

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Python | Machine Learning | Business Intelligence

June 2026